MLFF ADAPT-EVAL1 adaptive top-K full-evaluation specification

Status: implemented in mdstats 0.20.126a0.

Purpose

ADAPT-EVAL1 replaces the generated production EVAL-MF successive-halving workflow after ADAPT-MON1/STOP1/RANK1 have already paid for fixed epoch-wise target/replay monitor metrics. Production evaluation performs no 10% or 33% checkpoint rounds. It consumes exactly one immutable lightweight champion from each independent run, purchases authoritative full inference for a small campaign-wide finalist set, rejects hard failures before weighted ranking, and expands the purchased set only when the current batch contains no admissible model.

Historical `bounded`, `exhaustive`, and `multi_fidelity` evaluation strategies remain readable and executable for pre-adaptive campaign evidence. New generated campaigns use `checkpoint_strategy = "adaptive_topk"`.

Inputs and immutable lineage

ADAPT-EVAL1 requires:

- a completed training-campaign plan;
- terminal ADAPT-STOP1 state for every considered run;
- reconciled ADAPT-RANK1 `lightweight_run_champion.json` evidence;
- the frozen checkpoint catalog for each champion;
- one identical common full target artifact across all adaptive DATA8 bundles;
- one independent full TRUE_DFT replay artifact;
- the binary learned-model precision identity (`single` or `double`); and
- the retained full-evaluation safety policy.

The finalist queue is content-addressed to the campaign, the run-local champion evidence, and the scoring policy. Full candidate decisions are content-addressed to the corresponding checkpoint evaluation record.

Common full target domain

The 256-frame ADAPT-MON1 target monitor is a lightweight training/screening domain and must not be relabeled as full evidence. DATA8 v3 therefore materializes the complete DATA5 `outer_monitor` domain once as:

`shared/target/full_target_evaluation.xyz`

with role `common_full_target_evaluation`. The 256-frame online monitor must be a subset of this domain. All adaptive bundles competing in one campaign must expose identical full-target membership and content identity.

Common full replay domain

Authoritative replay evaluation uses the complete configured independent TRUE_DFT replay monitor domain resolved from `[paths].replay_true_labels` and the frozen replay-monitor lineage. The 512-frame online replay monitor remains a lightweight fixed trend monitor; it is not substituted for full replay evidence. This domain is the authoritative true-label replay validation/monitor split, not the replay-training corpus.

Both naive fine-tuning and multi-head replay candidates are evaluated on the same true-replay domain. A multi-head model uses its replay head. A naive model has no replay head, so its target head is evaluated on the replay structures. This gives one physically comparable replay force RMSE across training methods.

Lightweight cross-method comparability

The campaign finalist queue is meaningful only if ADAPT-RANK1 scores have the same target/replay semantics for every training method. Multi-head replay obtains its replay validation row from the normal `pt_head` loader.

Naive fine-tuning receives the same fixed 512-frame TRUE_DFT replay monitor as an auxiliary validation-only loader evaluated through its target head.

The auxiliary loader:

- contributes no gradients;
- does not add a trainable head;
- is inserted before the ordinary target loader, preserving MACE 0.3.16's historical last-validation-loader checkpoint/patience scalar as target-driven; and
- supplies the replay metric consumed by ADAPT-STOP1 and ADAPT-RANK1.

Thus all new adaptive runs rank by the same weighted target/true-replay force objective.

Finalist queue

For one champion per independent run, order candidates by the frozen lightweight score and its existing deterministic tie-breakers. Initially purchase full evaluation for at most:

```toml
[evaluation]
finalist_count = 5
```

If fewer than five champions exist, evaluate all of them.

The queue is fixed before authoritative inference begins; full-evaluation results do not reorder or mutate the queue.

Authoritative full score

For target force RMSE `T_full` and true-replay force RMSE `R_full`, define:

$$ S_{\rm full}

\frac{w_T T_{\rm full}+w_R R_{\rm full}}{w_T+w_R}. $$

The default score weights are 1:1.

The replay hard ceiling remains coupled to the target threshold and score weights:

$$R_{\max} = \frac{w_T}{w_R} T_{\max}.$$

With the default `T_max = 0.030 eV/A` and 1:1 weights, both target and replay hard ceilings are 30 meV/A.

Hard acceptance before ranking

A candidate is fully admissible only after it satisfies:

1. `T_full <= T_max`;
2. `R_full <= R_max`;
3. retained target energy-MAE safety limits;
4. retained focus/mobile-ion force-RMSE safety limits;
5. retained stress-RMSE safety limits;
6. retained worst-condition force-RMSE safety limits; and
7. required finite/complete metric evidence.

The weighted score is never allowed to compensate for a hard failure.

The historical foundation-relative replay degradation remains recorded as:

- foundation true-replay force RMSE;
- candidate minus foundation absolute degradation; and

- fractional degradation when the foundation baseline is positive.

It is diagnostic under the new default selector and is not the hard replay acceptance gate.

Rescue batching

If no candidate in the first purchased batch is admissible, purchase the next unevaluated champions in deterministic batches of:

```
[evaluation]
finalist_rescue_batch_size = 5
```

Continue only until:

- at least one purchased candidate is fully admissible; or
- the champion pool is exhausted.

Once a purchased batch produces an admissible candidate, production full evaluation stops. An exhaustive historical/reference evaluator may still be used explicitly for research, but it is not the generated adaptive production strategy.

Precision contract

The learned-model dtype is never normalized across finalists:

```
single -> FP32 model inference
double -> FP64 model inference
```

An FP32 checkpoint is not promoted to FP64 for evaluation, reconstruction, or export. mdstats converts model outputs to FP64 for SSE/RMSE accumulation, retained safety metrics, replay diagnostics, and weighted score calculation.

Restart and cache semantics

Full predictions are reusable only when all relevant identities match, including:

- run and checkpoint identity;
- evaluation policy identity;
- binary model dtype;
- common full target artifact identity and SHA-256;
- common full replay artifact identity and SHA-256; and
- reconstructed model/checkpoint lineage.

Interrupted evaluation resumes from authenticated completed predictions. It never repeats a completed finalist merely to rebuild aggregate evidence, and it never changes the frozen queue.

Outputs

ADAPT-EVAL1 persists:

- `CampaignFinalistQueueRecord`;
- one authoritative `CheckpointEvaluationRecord` per purchased finalist;
- one `FullEvaluationCandidateRecord` per purchased finalist;
- one aggregate `AdaptiveFullEvaluationRecord`; and
- `results/adaptive-full-evaluation.json` for a complete campaign.

The aggregate result orders the fully admissible candidates by full score, but ADAPT-EVAL1 does not perform deployment verification, fallback after verification failure, or final export. Those responsibilities remain ADAPT-VERIFY1.

Acceptance tests

ADAPT-EVAL1 is complete only when tests prove:

1. generated campaigns default to `adaptive_topk` and purchase no EVAL-MF partial rounds;
2. the initial queue contains at most five one-per-run champions;
3. rescue proceeds in batches of five only when no purchased candidate is admissible;
4. the 256-frame target online monitor is a subset of a separately materialized common full target domain;
5. the full true-replay domain is independent of the 512-frame online replay monitor;
6. naive and replay-trained runs expose comparable target+true-replay lightweight scores;
7. hard target/replay and retained safety gates execute before full-score ranking;
8. the old foundation-relative replay degradation is diagnostic rather than the default hard selector;
9. `single` and `double` preserve their own model inference dtype with FP64 scientific reductions;
10. interruption/restart reuses authenticated completed predictions; and
11. historical EVAL-MF policies remain readable and directly testable without remaining the generated default.